

emerging infectious disease

24 Nov 2025

emerging and re-emerging disease

Basically, anything we're worried about

- **emerging**: new to a host: most often (in humans) via *zoonotic spillover*, but could be an existing disease that is expanding its range/becoming more infectious, virulent, whatever
- *re-emerging*: a parasite that was previously thought to be controlled, but is of renewed concern. Due to socioeconomic change, landscape change, climate change, evolution of antibiotic/antiviral resistance, etc.)
- **encounter filter**: changing patterns of reservoir host/vector distribution (due to climate/landscape change), human contact, ...
- **compatibility filter**: changes via mutation, recombination, selection for resistance, ...
- **disease triangle** (from plant pathology): need co-occurring host, pathogen, environment (for transmission: also consider virulence, treatment resistance ...)
- Local change > global change
- Parasites may get better, or worse, or different due to changes ...

Malaria in North America

- established by immigration of enslaved people from west Africa
- landscape change (agricultural clearance/hydrological changes)
- well established by 1850s except in northern/high-altitude/arid places
- poorly diagnosed, haphazardly treated
- decline due to drainage 1900-1920
- 5000 deaths/year (1930s) -> <10 (1950s)
- improvements in nutrition, medical care, housing, insecticides

We now know that anopheline mosquito breeding throughout the country was sufficiently low so that it was difficult for overseas malaria to become established inside the country, and that the prognostication of a wide spread development of overseas malaria in the American anophelines was based on a failure to consider the very great reduction in mosquito breeding which had occurred since 1931.

Faust (1951)

R_0 for vector borne diseases = (R_0 vector-to-host $\times$ R_0 host-to-vector): proportional to mosquito-to-human ratio (but preventing bites reduces both components)

Malaria changes

Often driven by **local** landscape change:

- Combinations of hydrological cycle, temperature, predators: e.g. brick-making pits in western Kenya (J. C. Carlson et al., 2004)

- MacDonald & Mordecai (2019): deforestation increases malaria, but malaria decreases deforestation (statistical model, we don't know the mechanism: ecological change, increased human-vector contact?)

Mosquito-borne diseases in 'temperate' countries

- 'autochthonous'/'local transmission events'
- doesn't mean $R_0 > 1$, but does mean $R_0 > 0 \dots$
- local transmission of:
 - chikungunya (*Aedes albopictus*, *A. aegypti*: Europe, North America) [New York state, 2025] (Disease Control, 2025; Health, 2025)
 - Zika (Florida, Texas) (*Aedes*)
 - dengue (Argentina, California, Florida, Texas (Barrenechea et al., 2025; Centers for Disease Control, 2025)) (*Aedes*)
 - malaria (DeVita et al., 2025) (*Anopheles mosquito* spp)

Candida auris

- Casadevall et al. (2019); Casadevall et al. (2021); De Gaetano et al. (2024)
- newly emerging fungal pathogen (first isolated 2009)
- not found in fungal collections before 1996
- molecular clocks: disease-causing clades MRCA ~ 37 years ago
- multidrug resistant
- heat-tolerant (up to 42°, more than nearest relatives - fever-tolerant?)
- newly virulent? newly resistant? newly heat-tolerant?

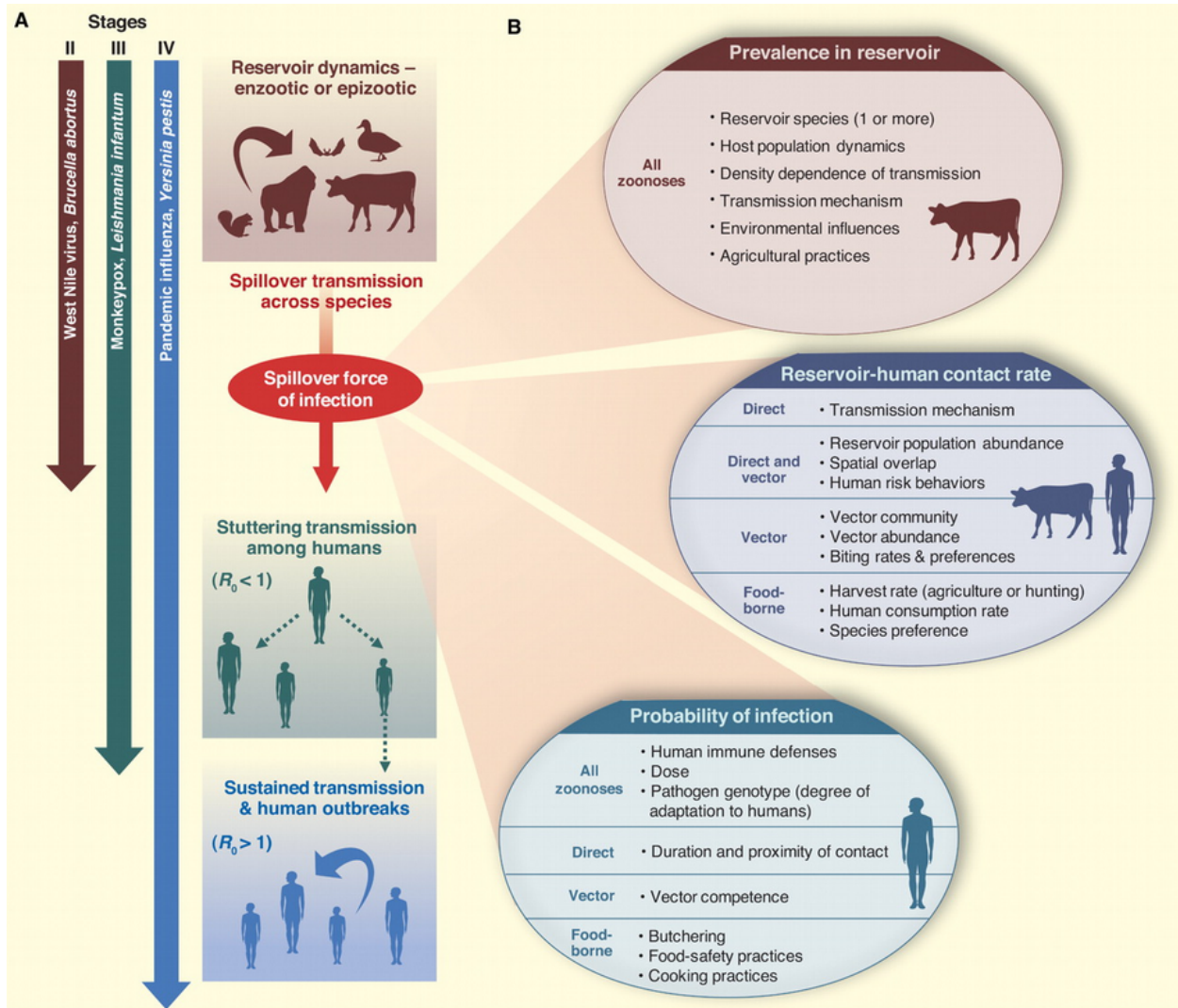
Batrachochytrium dendrobatidis

- fungal pathogen
 - most other chytrids are saprophytes, plant pathogens
 - *B. salamandrivorans*: salamander pathogen (more restricted)
- first discovered in poison dart frogs
- caused die-offs in E Australia, Central America, Colorado, California ...
- association with high altitude?

Very confusing ...

- declines occurred in pristine areas (probably not anthropogenic?)
- some species decline in the absence of Bd
- some species stable in the presence of Bd
- **tipping point hypothesis**: in populations all the time, but something happened to increase virulence/reduce tolerance or resistance ($\approx$ compatibility filter)
 - climate change/El Niño ?
 - ultraviolet radiation?
 - pesticides?
 - combination (species $\times$ temperature $\times$ U/V $\times$ pesticide $\times$...)? (Pounds et al., 2006; Rohr et al., 2008; Rohr & Raffel, 2010)
- **novel pathogen hypothesis**: mutation/speciation + dispersal
 - detection in historical specimens: CA/bullfrog, Brazil ...
 - genomics (challenging!)
 - Asian sampling

Fisher & Garner (2020)



theory

Do we need to understand *everything*?

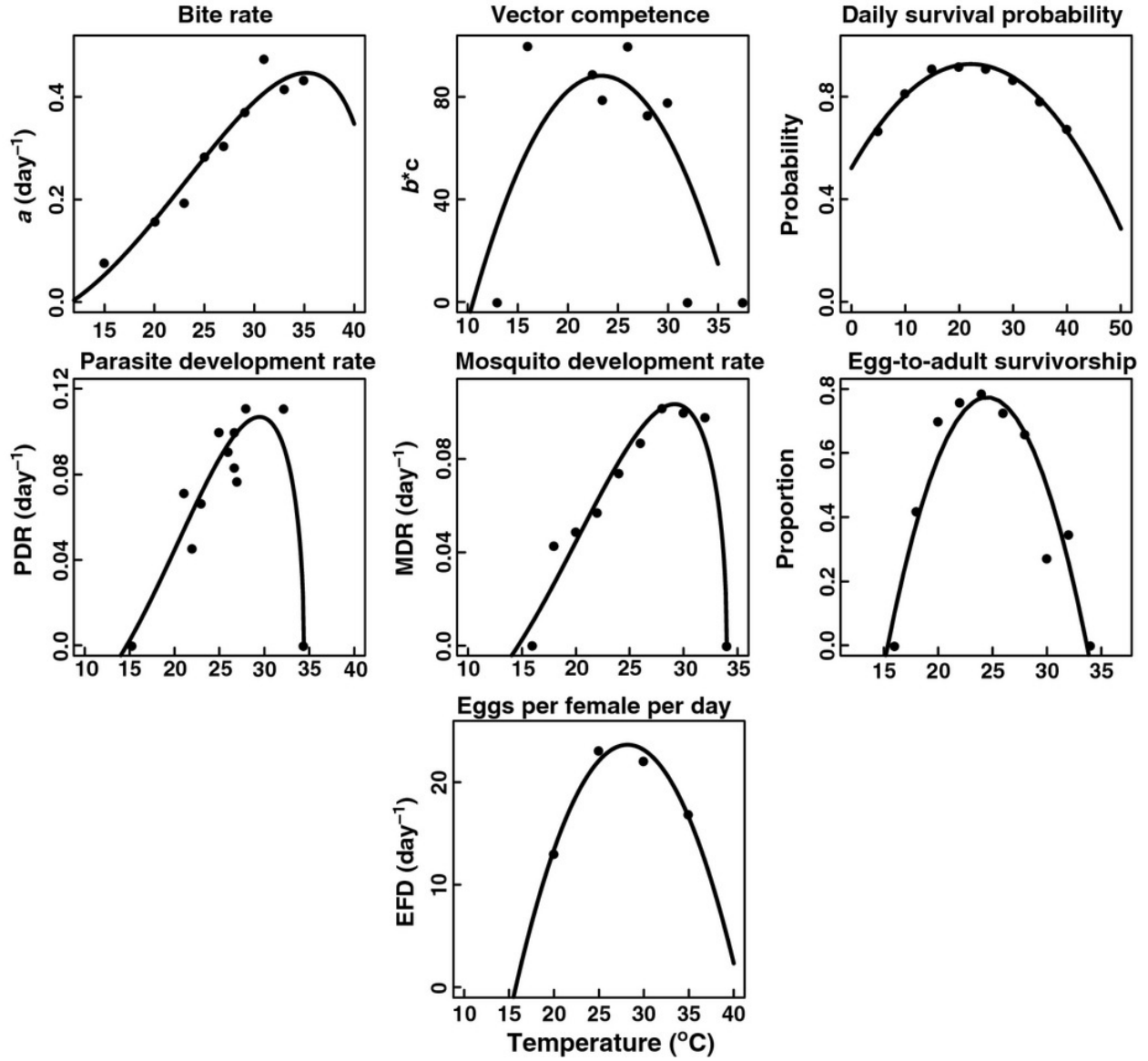
- reservoir ecology
- pathogen biology
- human-reservoir interactions

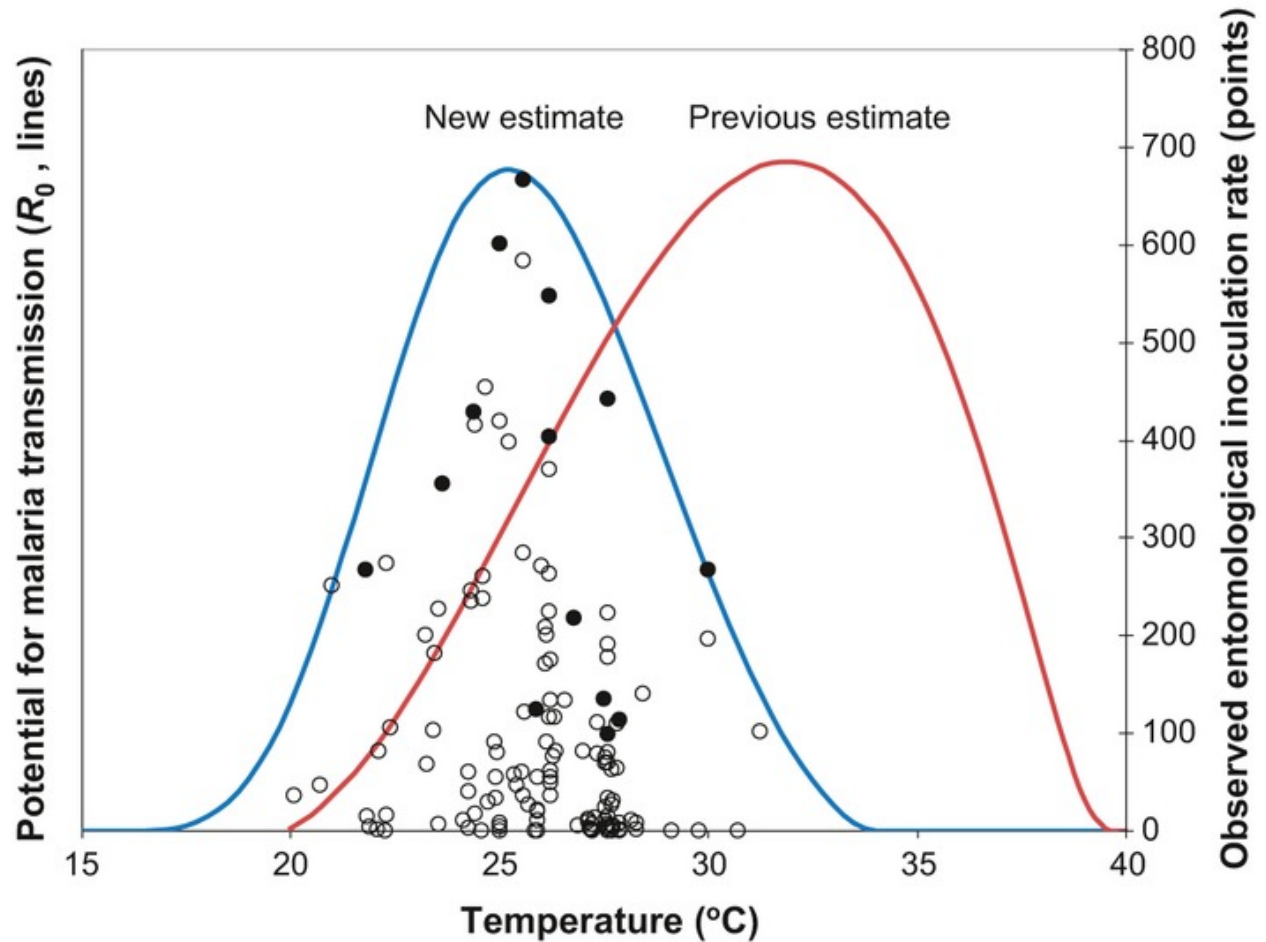
How do we understand? How do we predict?

effects of climate change

- warming
 - ‘good’ or ‘bad’ for pathogens?
 - vector biology
 - * extended range
 - * higher activity?
 - * lots of debate (Omumbo et al., 2011)
 - * details of temperature sensitivity: e.g. Mordecai et al. (2013); Ryan et al. (2015); Mordecai et al. (2020): shift from *Anopheles gambiae* to *Aedes aegypti*, malaria to arboviruses (dengue,

- chikungunya etc.);
- changes in seasonality, hydrological cycles
- local landscape change
 - hydrology
 - suburbanization and reforestation: Lyme disease
 - deforestation (MacDonald & Mordecai, 2019)
- changes in reservoir communities



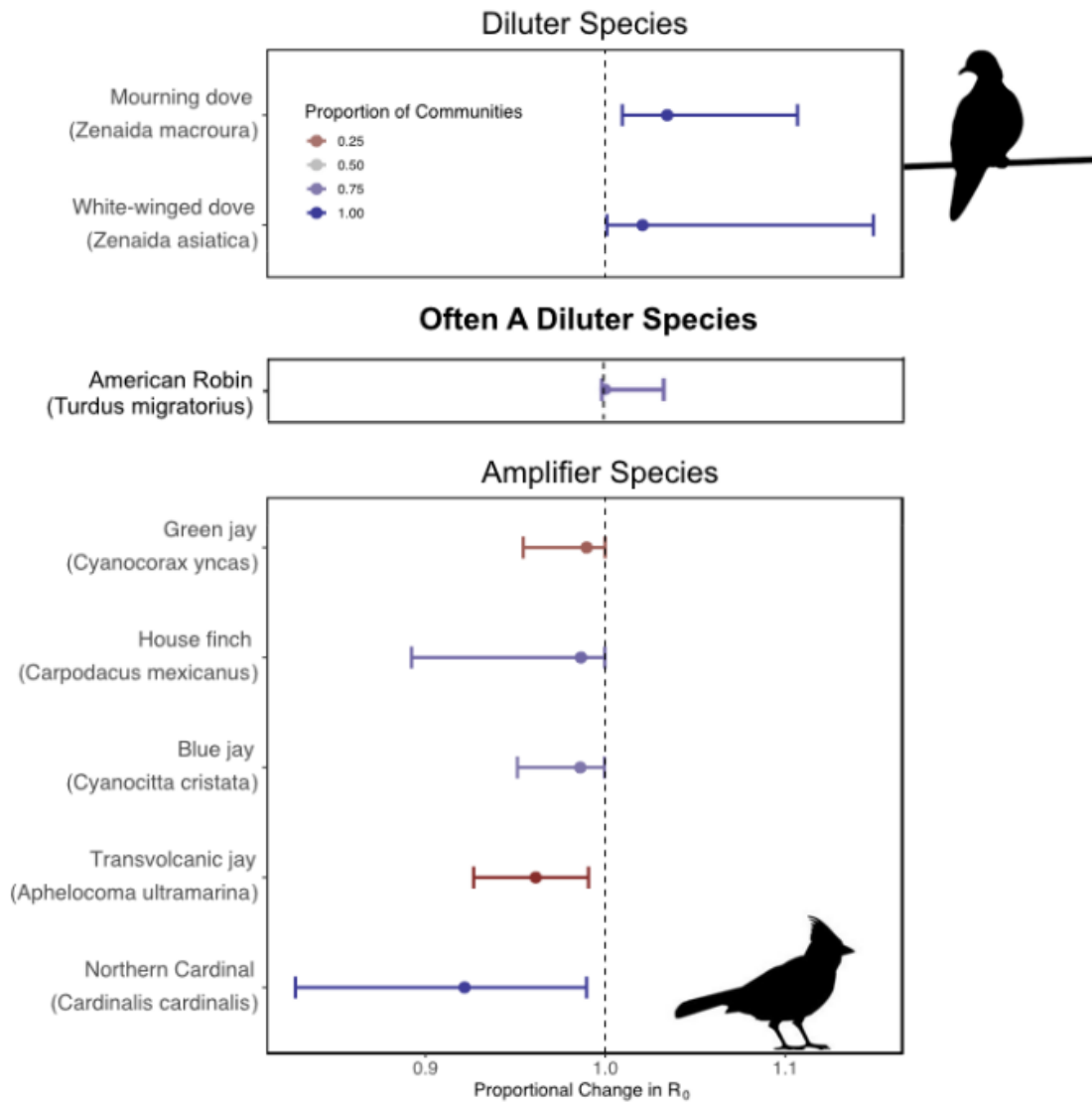


effects of biodiversity change: dilution effect (Keesing & Ostfeld, 2021)

- does increased biodiversity decrease disease?
- variation in reservoir competence
- high-quality hosts decrease with increasing biodiversity
 - encounter reduction; host regulation; vector preferences
- depends on how the community is changing
 - what is driving biodiversity loss?
 - what species are more likely to be lost?
 - are less/more **competent** host species more likely to be lost?

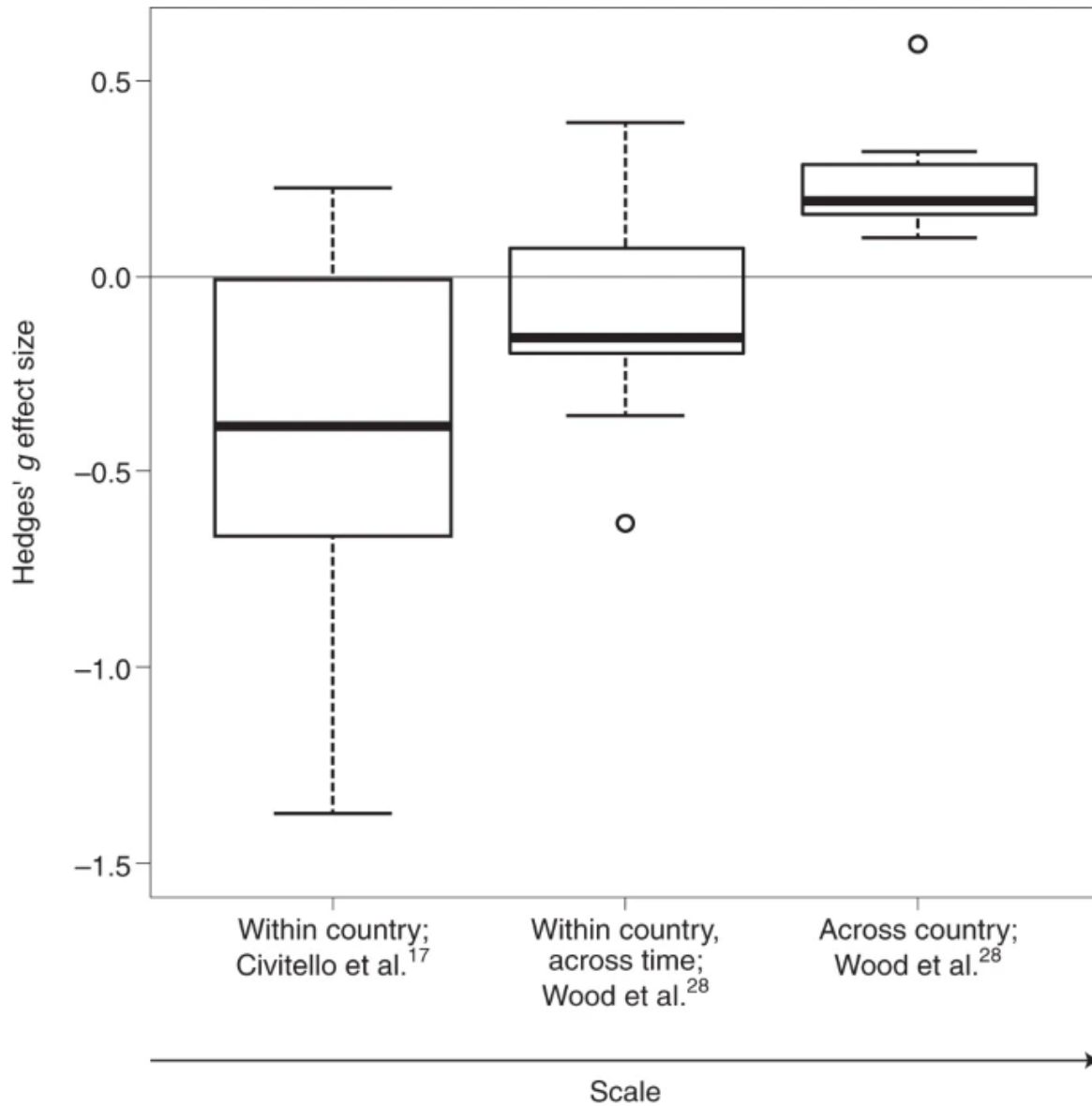
Kain & Bolker (2019):

species-level effects on WNV



Rohr et al. (2020): meta-analysis (again ...)

Fig. 4: Hedges' g effect sizes.



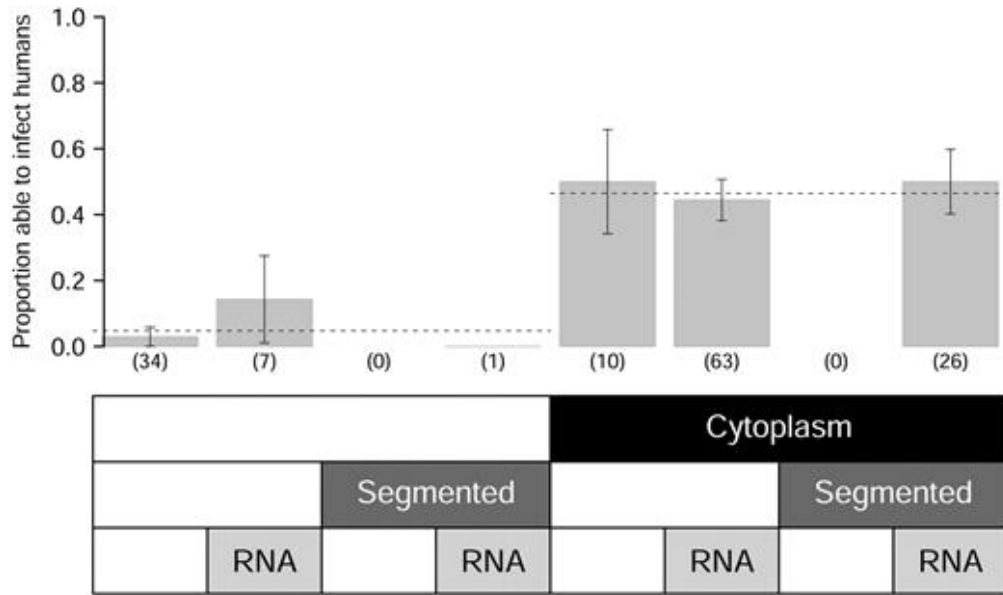
- C. J. Carlson et al. (2022): *higher* rodent diversity and climate anomalies drive plague spillover

surveillance and prediction

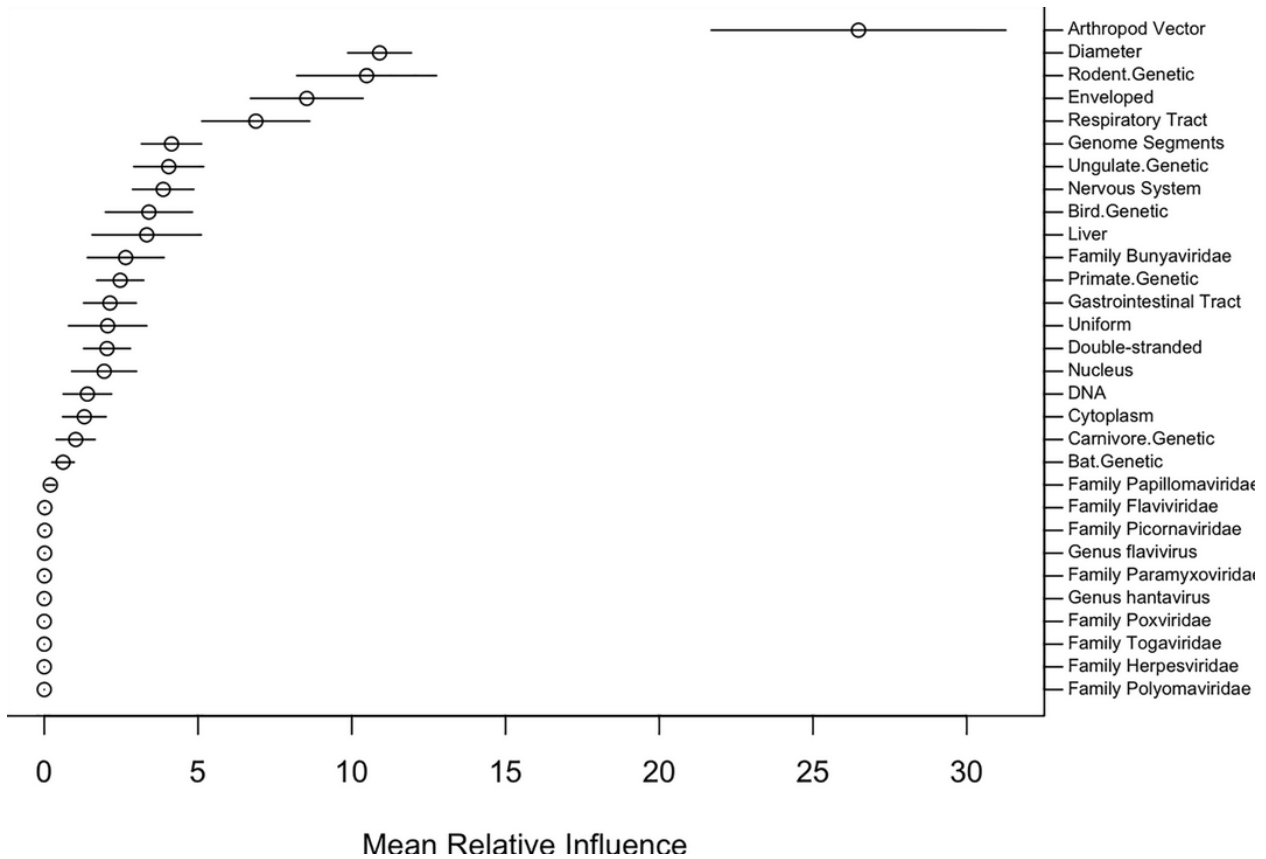
- which viruses will emerge, where, why?

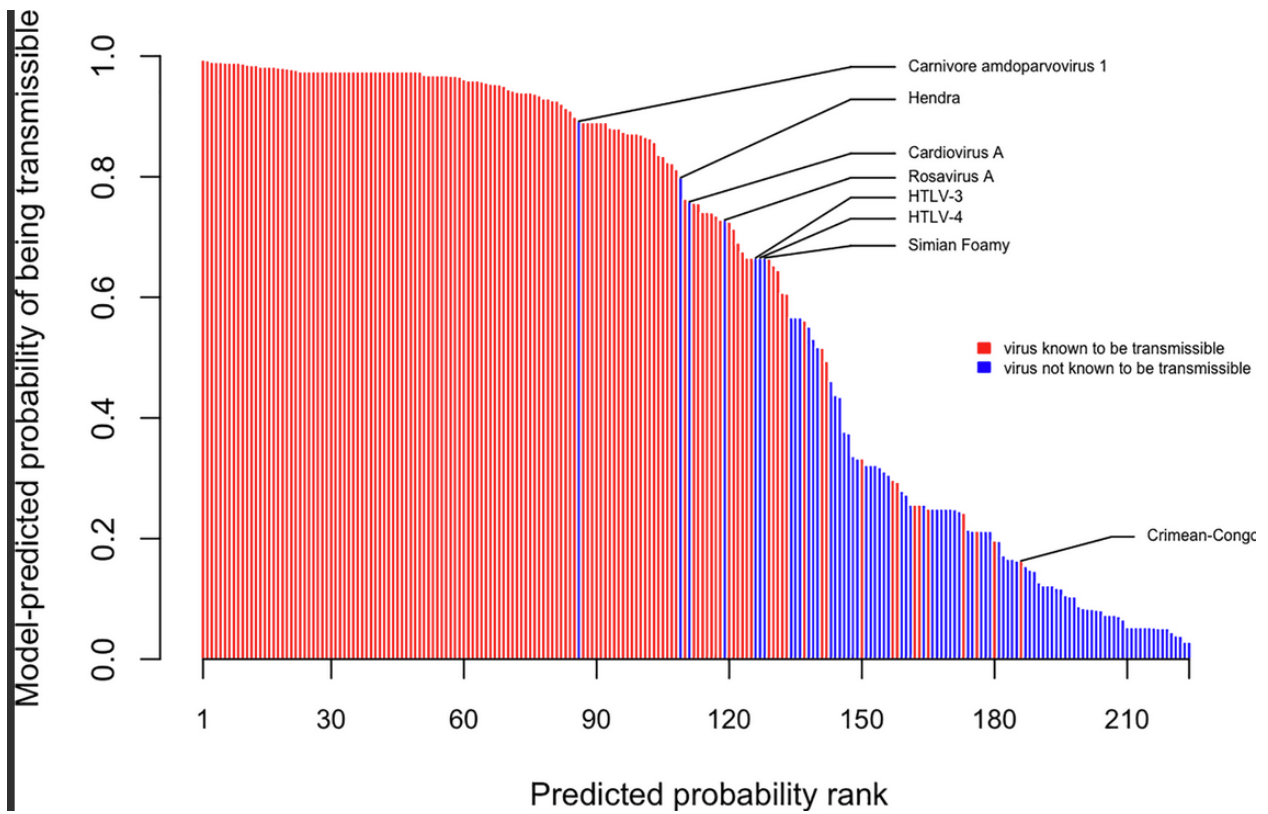
Pulliam & Dushoff (2009): predict zoonotic transmission of livestock viruses

Figure 1.



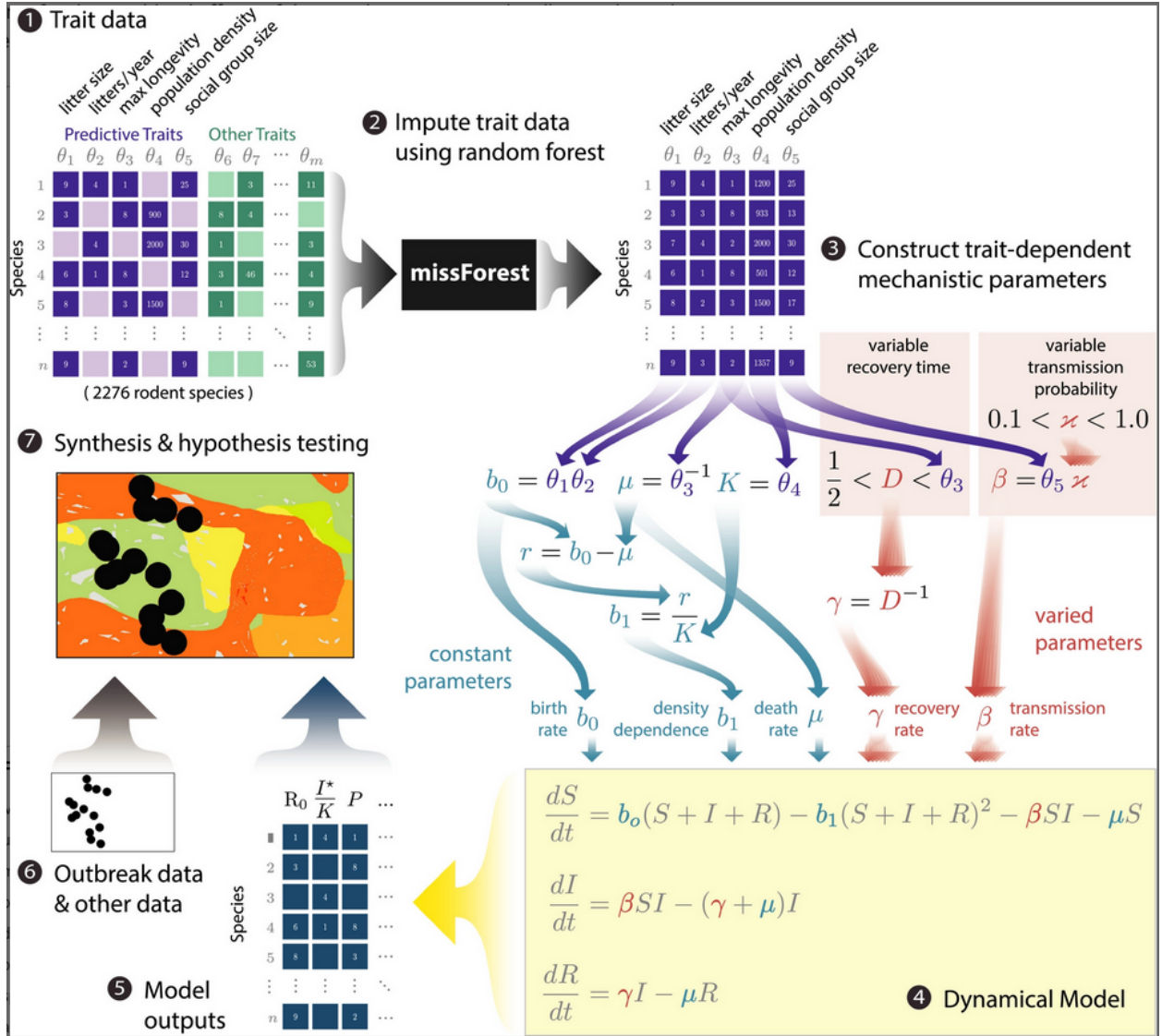
Walker et al. (2018): predict human transmission ability of zoonotic viruses

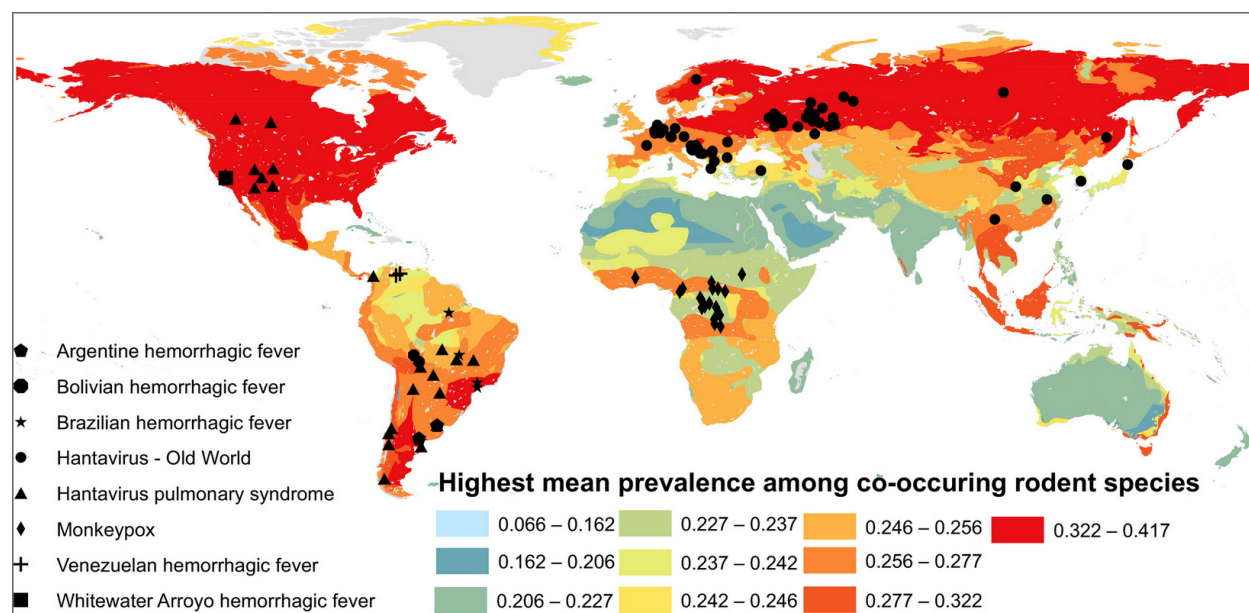




Evans et al. (2023): 2017-2020 sample: 12% of 693 individuals sampled in Myanmar were seropositive for sarbecovirus, more likely if they were loggers/hunters or had been exposed to bats ...

Han et al. (2020): model rodent life history





References

- Barrenechea, G. G., Sánchez, R., Calderón, O. C., Buffone, I. R., & Bastos, L. S. (2025). Characterization of the most austral autochthonous dengue outbreak reported in the world (city of Bahía Blanca, Argentina, January–June 2024). A cross-sectional study. *The Lancet Regional Health – Americas*, 51. <https://doi.org/10.1016/j.lana.2025.101254>
- Carlson, C. J., Bevens, S. N., & Schmid, B. V. (2022). Plague risk in the western United States over seven decades of environmental change. *Global Change Biology*, 28(3), 753–769. <https://doi.org/10.1111/gcb.15966>
- Carlson, J. C., Byrd, B. D., & Omlin, F. X. (2004). Field assessments in western Kenya link malaria vectors to environmentally disturbed habitats during the dry season. *BMC Public Health*, 4(1), 33. <https://doi.org/10.1186/1471-2458-4-33>
- Casadevall, A., Kontoyiannis, D. P., & Robert, V. (2019). On the Emergence of *Candida auris*: Climate Change, Azoles, Swamps, and Birds. *mBio*, 10(4), e01397–19. <https://doi.org/10.1128/mBio.01397-19>
- Casadevall, A., Kontoyiannis, D. P., & Robert, V. (2021). Environmental *Candida auris* and the Global Warming Emergence Hypothesis. *mBio*, 12(2), 10.1128/mbio.00360–21. <https://doi.org/10.1128/mbio.00360-21>
- Centers for Disease Control. (2025). Current Dengue Outbreak. In *Dengue*. <https://www.cdc.gov/dengue/outbreaks/2024/index.html>
- De Gaetano, S., Midiri, A., Mancuso, G., Avola, M. G., & Biondo, C. (2024). *Candida auris* Outbreaks: Current Status and Future Perspectives. *Microorganisms*, 12(5), 927. <https://doi.org/10.3390/microorganisms12050927>
- DeVita, T. N., Morrison, A. M., Stanek, D., Drennon, M., Sarney, E., Brennan, W., Tomson, K., Blackmore, C., Broussard, K., Duwell, M., Blythe, D., Rothfeldt, L., Dulski, T., Blount, K., Ledford, S., Blackburn, D., Wallender, E., Barratt, J. L. N., Raphael, B. H., ... McElroy, P. D. (2025). Public Health Response to the First Locally Acquired Malaria Outbreaks in the US in 20 Years. *JAMA Network Open*, 8(10), e2535719. <https://doi.org/10.1001/jamanetworkopen.2025.35719>
- Disease Control, C. for. (2025). Chikungunya in the United States. In *Chikungunya Virus*. <https://www.cdc.gov/chikungunya/data-maps/chikungunya-us.html>
- Evans, T. S., Tan, C. W., Aung, O., Phyu, S., Lin, H., Coffey, L. L., Toe, A. T., Aung, P., Aung, T. H., Aung, N. T., Weiss, C. M., Thant, K. Z., Htun, Z. T., Murray, S., Wang, L.-F., Johnson, C. K., & Thu, H. M. (2023). Exposure to diverse sarbecoviruses indicates frequent zoonotic spillover in human communities interacting with wildlife. *International Journal of Infectious Diseases*, 0(0). <https://doi.org/10.1016/j.ijid.2023.02.015>
- Faust, E. C. (1951). The history of malaria in the United States. *American Scientist*, 39(1), 121–130. <https://www.jstor.org/stable/27826354>

- Fisher, M. C., & Garner, T. W. J. (2020). Chytrid fungi and global amphibian declines. *Nature Reviews Microbiology*, 18(6), 332–343. <https://doi.org/10.1038/s41579-020-0335-x>
- Han, B. A., O'Regan, S. M., Paul Schmidt, J., & Drake, J. M. (2020). Integrating data mining and transmission theory in the ecology of infectious diseases. *Ecology Letters*, 23(8), 1178–1188. <https://doi.org/10.1111/ele.13520>
- Health, N. S. D. of. (2025). *New York State Department of Health Confirms First Locally Acquired Case of Chikungunya in New York State*. https://www.health.ny.gov/press/releases/2025/2025-10-14/_chikungunya
- Kain, M. P., & Bolker, B. M. (2019). Predicting West Nile virus transmission in North American bird communities using phylogenetic mixed effects models and eBird citizen science data. *Parasites & Vectors*, 12(1), 395. <https://doi.org/10.1186/s13071-019-3656-8>
- Keesing, F., & Ostfeld, R. S. (2021). Dilution effects in disease ecology. *Ecology Letters*, 24(11), 2490–2505. <https://doi.org/10.1111/ele.13875>
- MacDonald, A. J., & Mordecai, E. A. (2019). Amazon deforestation drives malaria transmission, and malaria burden reduces forest clearing. *Proceedings of the National Academy of Sciences*, 116(44), 22212–22218. <https://doi.org/10.1073/pnas.1905315116>
- Mordecai, E. A., Paaijmans, K. P., Johnson, L. R., Balzer, C., Ben-Horin, T., de Moor, E., McNally, A., Pawar, S., Ryan, S. J., Smith, T. C., & Lafferty, K. D. (2013). Optimal temperature for malaria transmission is dramatically lower than previously predicted. *Ecology Letters*, 16(1), 22–30. <https://doi.org/10.1111/ele.12015>
- Mordecai, E. A., Ryan, S. J., Caldwell, J. M., Shah, M. M., & LaBeaud, A. D. (2020). Climate change could shift disease burden from malaria to arboviruses in Africa. *The Lancet Planetary Health*, 4(9), e416–e423. [https://doi.org/10.1016/S2542-5196\(20\)30178-9](https://doi.org/10.1016/S2542-5196(20)30178-9)
- Omumbo, J. A., Lyon, B., Waweru, S. M., Connor, S. J., & Thomson, M. C. (2011). Raised temperatures over the Kericho tea estates: Revisiting the climate in the East African highlands malaria debate. *Malaria Journal*, 10(1), 12. <https://doi.org/10.1186/1475-2875-10-12>
- Pounds, A. J., Bustamante, M. R., Coloma, L. A., Consuegra, J. A., Fogden, M. P. L., Foster, P. N., La Marca, E., Masters, K. L., Merino-Viteri, A., Puschendorf, R., Ron, S. R., Sánchez-Azofeifa, G. A., Still, C. J., & Young, B. E. (2006). Widespread amphibian extinctions from epidemic disease driven by global warming. *Nature*, 439(7073), 161–167. <https://doi.org/10.1038/nature04246>
- Pulliam, J. R. C., & Dushoff, J. (2009). Ability to Replicate in the Cytoplasm Predicts Zoonotic Transmission of Livestock Viruses. *The Journal of Infectious Diseases*, 199(4), 565–568. <https://doi.org/10.1086/596510>
- Rohr, J. R., Civitello, D. J., Halliday, F. W., Hudson, P. J., Lafferty, K. D., Wood, C. L., & Mordecai, E. A. (2020). Towards common ground in the biodiversity–disease debate. *Nature Ecology & Evolution*, 4(1), 24–33. <https://doi.org/10.1038/s41559-019-1060-6>
- Rohr, J. R., & Raffel, T. R. (2010). Linking global climate and temperature variability to widespread amphibian declines putatively caused by disease. *Proceedings of the National Academy of Sciences*, 107(18), 8269–8274. <https://doi.org/10.1073/pnas.0912883107>
- Rohr, J. R., Raffel, T. R., Romansic, J. M., McCallum, H., & Hudson, P. J. (2008). Evaluating the links between climate, disease spread, and amphibian declines. *Proceedings of the National Academy of Sciences*, 105(45), 17436–17441. <https://doi.org/10.1073/pnas.0806368105>
- Ryan, S. J., McNally, A., Johnson, L. R., Mordecai, E. A., Ben-Horin, T., Paaijmans, K., & Lafferty, K. D. (2015). Mapping Physiological Suitability Limits for Malaria in Africa Under Climate Change. *Vector-Borne and Zoonotic Diseases*, 15(12), 718–725. <https://doi.org/10.1089/vbz.2015.1822>
- Walker, J. W., Han, B. A., Ott, I. M., & Drake, J. M. (2018). Transmissibility of emerging viral zoonoses. *PLOS ONE*, 13(11), e0206926. <https://doi.org/10.1371/journal.pone.0206926>

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